different aminoacyl – tRNA synthetases are known. The power to recognize codon on the mRNA lies in the tRNA and not in the attached amino acid molecule.

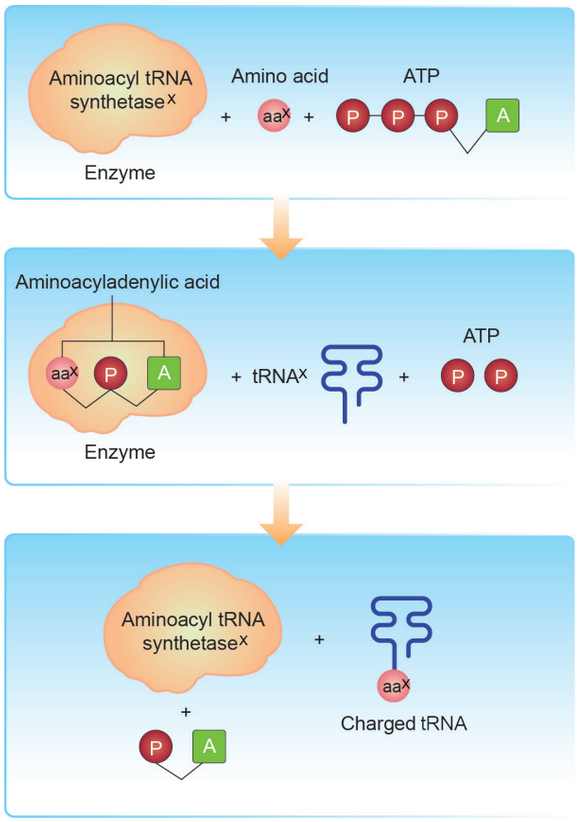


Fig. 5.12 Steps involved in charging tRNA. The 'X' denotes that for each amino acid only the corresponding specific tRNA and specific aminoacyl tRNA synthetase enzyme are involved in the charging process.

5.12 Translation

Translation refers to the process of polymerization of amino acids to form poly peptide chain. The decoding process is carried out by ribosomes that bind mRNA and charged tRNA molecules. The mRNA is translated, starting at the 5' end. After binding to mRNA, the ribosomes move along it, adding new amino acids to the growing polypeptide chain each time it reads a codon. Each codon is read by an anticodon on the corresponding tRNA. Hence the order and sequence of amino acids are defined by the sequence of bases in the mRNA.

5.12.1 Mechanism of Translation

The cellular factory responsible for synthesizing protein is the ribosome. The ribosome consists of structural RNAs and about 80 different proteins. In inactive state, it exists as two subunits; large subunit and small subunit. When the subunit encounters an mRNA, the process of translation of the mRNA to protein begins. The prokaryotic ribosome (70 S) consists of two subunits, the larger subunit (50 S) and smaller subunit (30 S). The ribosomes of eukaryotes (80 S) are larger, consisting of 60 S and 40 S sub units. 'S' denotes the sedimentation coefficient which is expressed as Svedberg unit (S).

One of the alternative ways of dividing up a sequence of bases in DNA or RNA into codons is called **reading frame**. Any sequence of DNA or RNA, beginning with a start codon and which can be translated into a protein is known as an **Open Reading Frame (ORF)**. A translational unit in mRNA is the sequence of RNA that is flanked by the start codon and the stop codon and codes for polypeptides. mRNA also have some additional sequences that are not translated and are referred to as **Untranslated Regions (UTR).** UTRs are present at both 5' end (before start codon) and at 3' end (after stop codon). The start codon **(AUG)** begins the coding sequence and is read by a special tRNA that carries methionine (met). The initiator tRNA charged with methionine binds to the AUG start codon. In prokaryotes, N - formyl methionine (f ^{met}) is attached to the initiator tRNA whereas in eukaryotes unmodified methionine is used. The 5' end of the mRNA of prokaryotes has a special sequence which precedes the initial AUG start codon of mRNA. This ribosome binding site is called the **Shine – Dalgarno sequence**

XII Std Biology-Zoology Chapter-5 EM.indd 75 23/12/2022 12:16:01